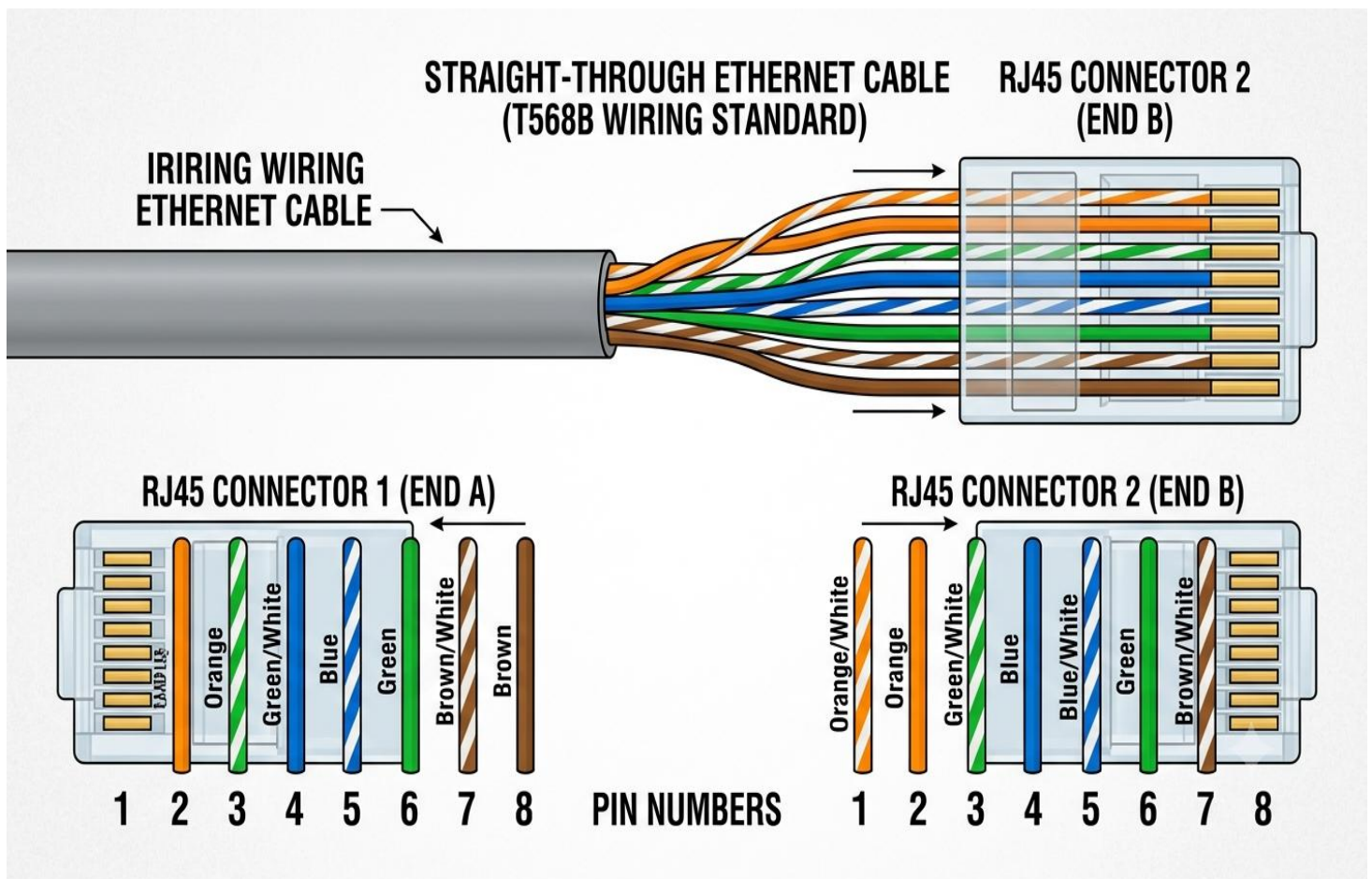


Network Cable and Connector

1. image of RJ45 connector :



2. Types of Network Cables & Real-World Applications :

Cable Type	Real-World Application / Device	Reason for Selection
Cat5e Cable	Zomato POS Terminal / Billing Machine	Cost-effective and handles standard 100 Mbps to 1 Gbps speeds over short distances, perfectly sufficient for low-data credit card processing and receipt printing.
Cat6 Cable	High-Speed Wi-Fi 6 Router / LAN PC	Supports up to 10 Gbps speeds with high frequency (250 MHz) and reduced crosstalk, making it ideal for streaming, online gaming, and fast local transfers.
Fiber Optic Cable	IPL Live Streaming / ISP Data Center Server	Transmits data using light pulses over long distances with zero electromagnetic interference, providing gigabit bandwidth needed for multi-million concurrent viewers.

3. Steps to Crimp an RJ-45 Connector :

- **Strip the Outer Jacket:** Use a cable stripper to remove roughly 1 inch of the plastic outer sheath from the end of the Ethernet cable, taking care not to nick the inner wires.
- **Untwist and Straighten Wires:** Separate the 4 twisted pairs and flatten out all 8 individual color-coded wires.

- **Arrange in T568B Order:** Line up the wires left-to-right: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
- **Trim the Wires:** Snip the aligned wires straight across using wire cutters, leaving about half an inch (12-13mm) exposed.
- **Insert into RJ-45 Plug:** Push the wires firmly into the RJ-45 connector until all 8 conductors reach the very tip and sit under the gold pins.
- **Crimp the Plug:** Place the RJ-45 connector into the crimping tool slot and squeeze the handles firmly until the gold pins clamp down onto the wire conductors.

4. Gaming Zone & Screening Setup Scenario :

Selected Equipment

- **Primary Cabling:** Cat6 Ethernet Cables
- **Connector Type:** RJ-45 Pass-Through Connectors

Reasoning

- **Low Latency & High Speed:** Gaming consoles demand consistent sub-15ms ping and zero packet loss. Cat6 cables handle 1 Gbps to 10 Gbps speeds reliably without wireless lag.
- **Interference Resistance:** Cat6 cables feature a central plastic spline (separator) that minimizes signal crosstalk, which is essential when dozens of network lines run parallel to heavy power cables feeding large screens and consoles.

